DRAGSTER DESIGN

OVERVIEW

Applying leadership and 21st century skills, participants design, produce a working drawing for, and build a CO_2-powered dragster according to stated specifications, using only specified materials. Special annual design requirements (if any) will be posted for this event on the TSA website under *Themes & Problems*.

ELIGIBILITY

Two (2) individuals per chapter may participate.

TIME LIMITS

A. Dragster and documentation portfolio are submitted at the time and place stated in the conference program.

B. All raceable cars will make one (1) qualifying time run, then judges check for spec compliance, leaving the top sixteen (16) legal cars for the Final Race.

C. Sixteen (16) qualifying car builders participate in a five (5)-minute interview.

D. Documentation portfolios and cars must be picked up at the specified time and place stated in the conference program.

ATTIRE

TSA competition attire is required for this event.

PROCEDURE

PRE-CONFERENCE

A. Participants review the TSA Honor Statement for Competitive Events found in the General Rules.

B. Participants review the Use of Generative Artificial Intelligence (GenAI) found in the General Rules.

C. Participants prepare their documentation and Dragster model according to the regulations.

PRELIMINARY ROUND

A. Participants report at the time and location stated in the conference program to check in:

1. Documentation portfolio
2. Dragster

B. Entries are reviewed by judges to determine safety on the track.

C. Safe dragsters race for qualifying time on the same lane of the raceway.

D. Entries are evaluated by the judges with neither students nor advisors present based on the following criteria:

1. Judges determine the top sixteen (16) qualifying entries based on the time trials and event criteria to determine the semifinalists.
2. Judges score the dragster construction, interview, and race points to determine the top ten (10) finalists.

E. Dragsters that do not meet event regulations are disqualified and lower qualifying cars are moved up until sixteen (16) dragsters that meet specifications are determined.

F. Following the completion of the time trials, race times are posted at the event location for the duration of the semifinal round.

G. A semifinalist list (in random order) is posted.

SEMIFINAL ROUND

A. The top sixteen (16) dragster builders report to the track at the posted time for a five (5)-minute interview.

B. The top sixteen (16) entries race in a double-elimination format to earn points for the race portion of the event. An optional single lane head-to-head race may be used as determined by the TSA CRC.

C. Drawing, design, body finish, parts list, references and AI reflection, and semifinal interview points are combined with race points to determine the final standings.

D. Following the race, participants pick up their entries from the display area at the time and place stated in the conference program.

E. The top ten (10) finalists are announced at the awards ceremony.

Rule DB. Dragster Body	MINIMUM	MAXIMUM
1. One (1) piece of any species of wood or from plastics, including urethane modeling foam. May be produced using traditional hand tools, power tool, CNC Milling and 3D printing, following applicable safety rules. a. Two (2) or more like or unlike pieces of material glued together are not considered one (1)-piece b. Any type of lamination will result in disqualification. c. No add-ons, such as body strengtheners, fenders, plastic canopy, exhausts, or air foils may be attached to or enclosed within the vehicle. Hydro dipping technique is permitted. d. Fiberglass, vinyl wrap, and shrink wrap are considered body strengtheners and cannot be used on the car body for any reason. e. Decals may be used for decoration only; they may not be used to gain an aerodynamic advantage, i.e., decals cannot cover the exterior axle holes or be used to cover open areas of the body.		
2. Body length	(2027) 265mm (2028) 290mm	(2027) 275mm (2028) 300mm
3. Body height with wheels		75mm
4. Body mass (completed car without CO_2)	(2027) 60g (2028) 85g	(2027) N/A (2028) N/A
5. Body width at the point the axles pass through the body, front and back	35mm	42mm
6. Vehicle total width (including wheels).		90mm

Rule A. Axle Holes/Wheelbase	MINIMUM	MAXIMUM
1. Dragsters must have two (2) axles per car, no more.		
2. Bottom of axle hole or bearing above bottom of car body. (NOTE: This will only be measured at the side surfaces of the car body at the axle holes. Measurement will made at the Rule DB: Dragster Body #5 location.)	5mm	10mm
3. Axle hole from front and rear of car	15mm	100mm
4. Wheelbase (axle distance apart at farthest points)	105mm	Not Specified
5. Bearings, bushings and lubricants may be used.		
6. Glue may be used to secure bearings.		

Rule S. Spacer Washers/Clips	MINIMUM	MAXIMUM
1. Spacer washers		10
2. Axle clips		8
3. Silicone or any other type of glue/adhesive may not be used in place of wheel clips to hold wheels or axles in place.		

Rule P. Power Plant (CO_2 Cartridge Hole)		
	MINIMUM	MAXIMUM
1. The power plant hole must be at the farthest point at the rear of the car and must be parallel to the racing surface to assure proper puncture of the CO_2 cartridge. A minimum of 5mm thickness around the entire power plant hole must be maintained on the dragster for safety. The inside of the power plant hole must not be intentionally painted.		
2. Hole depth	45mm	55mm
3. Safety zone thickness	5mm	
4. Chamber diameter	19mm	20mm
5. Lowest point of chamber diameter to race surface (with wheels)	26mm	40mm

Rule E. Screw Eyes/Eyelets		
	MINIMUM	MAXIMUM
1. Dragsters must have two (2) screw eyes or eyelets (no more). Screw eyes/eyelets must not make contact with the racing surface. The track string must pass through both screw eyes/eyelets, which are located on the center line of the bottom of the car. Glue may be used to reinforce the screw eyes. It is the responsibility of the car designer/engineer to see that the screw eyes/eyelet holes are tightly closed to prevent the track string from slipping out. As with all adjustments, this must be done prior to event check-in.		
2. Inside diameter	3mm	5mm
3. Minimum distance apart (at farthest points)	150mm	N/A

Rule W. Wheels		
	MINIMUM	MAXIMUM
1. A dragster must have four (4) wheels, no more. a. Two (2) wheels must meet the requirements in #2 and #3 below. b. The other two (2) wheels must meet the requirements in #4 and #5 below. c. All four (4) wheels must touch the racing surface at the same time. d. All wheels must roll. e. Wheels must be made entirely from plastic. f. Dimensions must be consistent for the full circumference of each wheel. g. Measurement represents the FULL surface contact point where wheel makes contact with the track.		
2. Front diameter	30mm	40mm
3. Front width (at full, unbroken, surface contact point)	1.5mm	5mm
4. Rear diameter	35mm	40mm
5. Rear width (at full, unbroken, surface contact point)	12mm	18mm

REGULATIONS AND REQUIREMENTS

Students will work to develop their leadership and 21st century skills in the process of preparing for and participating in this TSA competitive event. The development and application of those skills must be evident in their submission, demonstration, and/or communication pertaining to the entry.

PRELIMINARY ROUND

A. Participants must check for "Special Design Challenge Requirements" for the current year's design challenge specifications, if any, on the TSA website under *Themes & Problems*.

B. Documentation Portfolio

1. Documentation materials (comprising "a portfolio") are required and must be secured in a clear front report cover with the following single-sided, 8½" x 11" pages, in this order:
    a. Title page with the event title, participant identification number, the conference city and state, and the year; one (1) page
    b. Table of contents; one (1) page
    c. Technical Drawing; one (1) page
        i. The two (2)-view (top and side) drawing of the completed vehicle, with metric dimensions, is made on one (1) sheet of paper no larger than 11" x 17" in size. Drawing may be folded as necessary to fit in the portfolio.
        ii. Technical drawing is developed using standard engineering practices and procedures.
        iii. Technical drawing may be produced using traditional drafting methods or CAD.
        iv. Title block includes only the participant's identification number, which is assigned at registration time and is placed on the entry and drawing during check-in.
        v. Technical drawing dimensions should be determined from the rules/regulations. See rules DB, A, S, P, E, and W.
    d. Materials List; a detailed list of all equipment, supplies, and materials used in creating the entry and technical drawing one (1) page
    e. Resources and AI Reflection (see 2027 & 2028 Forms Appendix); maximum of four (4) pages

C. Dragsters that do not meet the specifications/ tolerances are disqualified from the race.

SEMIFINAL ROUND

A. Race:

1. Official distance between the start line and the finish line on the race track is twenty (20) meters.
2. No repair or maintenance is allowed after the entries have been registered.
3. Any entry damaged during the race is evaluated by the event coordinator to determine whether or not the vehicle is allowed to race again.
4. In the event that the vehicle is damaged by conference personnel, the event coordinator rules as to whether or not the vehicle may be repaired by the student entering the vehicle. This is the only reason a student is allowed to touch his/her vehicle after registration.
5. Cars that lose wheels, bearings, screw-eyes will not continue to race.
6. Damaged wheels may not be replaced.
7. All CO_2 cartridges for the race are provided by national TSA.

B. Semifinal interview times require the competitor to sign up for a time slot after the top 16 cars are posted.

EVALUATION

PRELIMINARY ROUND

Tier 1

A. Sixteen (16) qualifying entries

B. Documentation portfolio

SEMIFINAL ROUND

Tier 2

C. Dragster construction, interview, and race points

Refer to the official rating form for more information.

GENERATIVE ARTIFICIAL INTELLIGENCE

Generative Artificial Intelligence (GenAI) is permitted for this event. Refer to the Use of Generative Artificial Intelligence (GenAI) for Competitive Events found in the General Rules and Regulations.

STEM INTEGRATION

This event aligns with the STEM (Science, Technology, Engineering, and Mathematics) educational standards.

LEADERSHIP AND 21ST CENTURY SKILLS

This event provides opportunity for students to build and develop leadership and 21st century skills including but not limited to – Communication, Collaboration/Social Skills, Initiative, Problem Solving/Risk Taking, Critical Thinking, Perseverance/Grit, Creativity, Relationship Building/Teamwork, Dependability/Integrity, and Flexibility/Adaptability

CAREERS RELATED TO THIS EVENT

This competition has connections to one (1) or more of the careers below:

- Aeronautical engineer
- Automotive designer
- Automotive modeler
- Industrial designer
- Industrial engineer
- Mechanical engineer
- Race car engineer

Participant/Team ID# _____

DRAGSTER DESIGN

2027 & 2028 OFFICIAL RATING FORM

HIGH SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ TIER 1 – Dragster is present
☐ TIER 1 – Documentation portfolio is present
☐ TIER 2 – Dragster construction, documentation portfolio, interview, and race points
☐ ENTRY NOT EVALUATED

PRELIMINARY DRAGSTER QUALIFYING RACE / POST RACE INSPECTION	YES	NO
Qualified Top 16 in Qualifying Race Speed		
Passed Top 16 Regulations Inspection		
Qualified Top 16 Legal (still safe to race) Cars for Interview and Final Race		
Please mark an "X" for each criteria. If all three are marked YES, place an "X" in the final box.		

SEMIFINAL DRAGSTER CONSTRUCTION (30 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Dragster Body Production Quality (X1)	Dragster exhibits poor production quality, with a crude and rough surface and little or no attention to detail.	Dragster shows evidence of proper production techniques; it is adequate but may need improvement.	Dragster displays excellent production techniques, with obvious attention to detail and quality.	
Body Paint/Finish (X1)	Surface preparation is inadequate; body is unprimed, with poorly applied final finish.	Dragster body is painted and finished adequately.	Dragster body finish is exemplary; body is smooth, shiny, and exhibits quality.	
Vehicle Assembly (X1)	Dragster exhibits poor or sloppy assembly of parts (loose wheels, eye screws are not level, and/or they are loose, etc.).	Dragster is well assembled, and adequately meets standards.	Dragster is properly assembled, with obvious evidence of attention to detail.	
			SEMIFINAL DRAGSTER CONSTRUCTION SUBTOTAL (30 points)	

SEMIFINAL DOCUMENTATION PORTFOLIO (50 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Portfolio Contents (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio has most components and is generally organized per the order of the event guide listing; it has sufficient content.	All components are included in the portfolio; content and organization per the order of the event guide listing are excellent.	
Drawing Scale and Dimensioning (X1)	Drawing is present, but is not to scale; dimensions are missing, or dimensioning is poorly done.	Drawing is acceptable and to scale; it is a close representation of the vehicle, but some dimensions may be missing.	Drawing is exemplary, exact, and to scale; includes all parts of the vehicle, and includes all dimensions in the regulations.	
Drawing Completion and Quality (X1)	Drawing is sloppy, missing parts, and lacking quality.	Drawing is complete, and the quality is adequate.	Drawing is complete and precise, and of exceptional quality.	
Materials List (X1)	Materials List is present, but has very few items listed and lacks organization; has the appearance of being thrown together at the last second.	Materials list is present and lists the majority of parts.	Materials list is complete and detailed.	
Resources and AI Reflection (X1)	Form is incomplete or vague and does not clearly explain whether or how AI was used, showing little connection to the submission or understanding of effectiveness and ethics.	Form is complete and clearly explains AI use or the lack thereof with connections to the submission or theme/topic (when applicable), addressing effectiveness and ethical considerations with adequate analysis.	Form is complete, concise, and specific, critically evaluating how AI or non-AI tools shaped decisions and outcomes, with strong alignment to the submission or theme/topic (when applicable) and ethical reasoning.	
			SEMIFINAL DOCUMENTATION PORTFOLIO SUBTOTAL (50 points)	

SUBTOTAL (80 points)	

SEMIFINAL INTERVIEW (20 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Car Builder Interview (X2)	Participant shows very limited knowledge of (and has difficulty articulating) how the car was produced or decisions made during the production; participant exhibits a basic understanding of design elements and functionality, and the rationale is inconsistent or absent.; leadership and/or 21st century skills are not evident.	Participant demonstrates some knowledge of the dragster production and has adequate knowledge of some processes or reasoning behind the vehicle design; leadership and/or 21st century skills are somewhat evident.	Participant demonstrates competence and knowledge related to the design and production of the dragster and articulates the "reasoning" behind the decisions made; leadership and/or 21st century skills are clearly evident.	
			SEMIFINAL INTERVIEW SUBTOTAL (20 points)	

SEMIFINAL RACE (55 points)								
1st	2nd	3rd	4th	5th & 6th	7th & 8th	9th-12th	13th – 16th	
55 Points	50 Points	45 Points	40 Points	35 Points	30 Points	25 Points	15 Points	
							SEMIFINAL RACE SUBTOTAL (55 points)	

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary. **TOTAL (155 points)**

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____

DRAGSTER DESIGN
EVENT COORDINATOR INSTRUCTIONS

PERSONNEL

A. Event coordinator
B. Judges, two (2) or more
C. Recorder for double elimination chart, one (1)
D. Assistants, two (2)

MATERIALS

A. Coordinator's packet and box, containing:
   1. Event guidelines, one (1) copy for the coordinator and for each judge
   2. TSA Event Coordinator Report
   3. Stick-on labels for entries, as needed
B. Time trial record sheet
C. Interview questions
D. Qualifier interview time slot sheet
E. Double elimination bracket chart
F. CO_2 cartridges
G. Metric scientific scales (triple beam balance or digital)
H. Mono-filament fishing line (suggest between 30 and 50 pound); four (4) pre-tied: two (2) on track and two (2) reserve, for the track
I. Race track set, including a starting gate and a finish gate, with a digital timer and winning lane indicator
J. Padding for the finish gate
K. Extra vehicles to test and demonstrate the track
L. Race brackets for placement of the semifinalists
M. Tables for the display of cars and for evaluation
N. Table at the starting line, for arranging and holding cars prior to the races
O. Table at the finish gate for the placement of cars after the races and to hold eliminated cars
P. Table for the official timekeeper
Q. When using a computer controlled track, provide the proper computer for the software being used, all necessary connections, and a printer. This equipment is placed on the official timekeeper's table.
R. Provide for a display of time trials and race brackets.

RESPONSIBLITIES

AT THE CONFERENCE

A. Attend the mandatory coordinator's meeting at the designated time and location.
B. Report to the CRC room and check the contents of the coordinator's packet.
C. Review the event guidelines and check to see that enough personnel have been scheduled.
D. Inspect the area(s) in which the event is to be held for appropriate set-up, including room size, tables, chairs, etc. Notify the event manager of any potential problems.
E. At least one (1) hour before the event is to begin, meet with judges and assistants to review time limits, procedures, regulations, evaluation, and all other details related to the event. If questions arise that cannot be answered, speak to the event manager before the event begins.
F. Ensure the judges have access to the online judging system.

EVENT CHECK-IN

A. Participants report to the time and place stated in the conference program and check in:
   1. Dragster
   2. Documentation portfolio
B. Late entries are considered on a case-by-case basis and only when the lateness is caused by events beyond the participant's control.
C. In order to compete, participants must be on the entry list or must have approval of the CRC.
D. If check-in is on the first day of the conference, requirements for attire do NOT apply during check-in.
E. Check to see that each entry drawing includes the participant's identification number in the upper right-hand corner of the paper.
F. Position each entry (dragster and drawing) for evaluation and viewing.
G. Secure the entries in the designated area.

PRELIMINARY ROUND

A. Begin the time trials at the scheduled time.
   1. Every race-worthy car should be tested.
   2. Students do not have to be present.
   3. Public viewing is allowed.
   4. Each car is timed in the same lane.
   5. Cars are timed only once.
   6. It is important that each car be positioned as well as possible in the starting gate.

B. Position a judge at the starting gate to ensure that all cars are positioned in the starting gate correctly.

C. Position another judge at the finish line.

D. If there is a misfire or if a time is not properly recorded, a rerun may be ordered at the discretion of the event coordinator.

E. Record preliminary times on a time trial record sheet.

F. Judges independently assess the entries using the following procedure:
   1. Judges determine the top sixteen (16) qualifying entries based on the time trials and event criteria. Note: When evaluating vehicles, judges are not allowed to disassemble or adjust vehicles. Dragsters evaluated/measured as-is.

G. Entries that do not meet specifications are removed.

H. Cars that are damaged or broken during the qualifying round are deemed non-raceable and also are removed.

I. Only raceable cars, as determined by the judges, are allowed to compete for the semifinalist category.

J. Lower qualifying cars are moved up until there are sixteen (16) legal semifinalists.

K. Decisions about rules violations must be discussed and verified with the judges, event coordinator, and CRC manager to determine either:
   1. To deduct twenty percent (20%) of the total possible points in this round or
   2. To disqualify the entry

   The event coordinator, judges, and CRC manager must initial either of these actions on the rating form.

L. Place each car in the double elimination race bracket (see sample) according to the rank of its qualifying time.

SEMIFINAL ROUND

A. Assist judges with the evaluation of dragster design, construction, and drawing.

B. Post the top sixteen (16) cars with interviews times.

C. Car builders report to the track at the posted time for a five (5)-minute car builder interview.

D. Conduct interviews with the qualifying top sixteen (16) car builders.

E. Begin the semifinals at the scheduled time.

F. Run the semifinalist race. A sample double-elimination bracket follows.

G. Only the sixteen (16) qualifying cars are raced.

H. Students do not have to be present.

I. Public viewing is allowed.

J. Judges independently assess the entries using the following procedure:
   1. Judges score the dragster construction, interview, and race points to determine the top ten (10) finalists.Judges use qualifying times to break any ties among the sixteen (16) qualifying cars.

K. If necessary, manage the security and removal of materials from the event area.

RACE BRACKET FOR 16-CAR DOUBLE ELIMINATION

